

Below you will find a curated list of sample questions covering the Multimedia Communication course.

To help you understand the level of detail, technical rigor, and analytical depth expected in your exam, I have provided a complete example answer for some representative questions. These examples illustrate how you should structure your responses, properly define theoretical concepts, and evaluate the engineering trade-offs between different coding algorithms.

I strongly encourage you to answer the remaining questions autonomously by relying on the lecture slides, your notes, and the reference textbook. Active synthesis of the material is the best way to master the subject.

Please keep in mind that this list is strictly indicative and not exhaustive. The actual written examination will feature different theoretical questions and **numerical exercises** designed to test your comprehensive understanding of the course content, and it will include topics not present in this document.

Audio and Speech Compression

Open-Ended Questions

1. Explain the difference between Source-Based (Parametric) coding and Sink-Based (Perceptual) coding.

2. Describe the “Analysis-by-Synthesis” (AbS) loop used in CELP (Code Excited Linear Prediction) codecs and why it represents an improvement over simple LPC-10.

Answer: LPC-10 uses a simplified excitation model (impulse train or white noise) that often results in unnatural-sounding speech. CELP improves this by using a codebook-based excitation (not just periodic pulses and white noise), introduces a closed loop search, and uses perceptual quality. The AbS loop functions as a “mini-decoder” inside the encoder: for each potential excitation vector from the codebook, the encoder synthesizes the speech and compares it against the original input. It then selects the vector that minimizes the perceptually weighted error, ensuring the chosen excitation yields the highest quality reconstruction, which is significantly more flexible and robust than the static LPC-10 excitation.

3. What is the role of the psychoacoustic masking model in perceptual audio coding, and how is it used to allocate bits?

4. Describe the principles of the LPC10 speech coding scheme

5. Draw the scheme and describe the operation of the functional blocks of an MP3 encoder

Multiple Choice Questions

1. Why are Line Spectrum Frequencies (LSF) preferred over direct quantization of LPC coefficients (a_i)?

- A) LSFs guarantee filter stability because their roots always lie on the unit circle and possess an interlacing property that can be easily verified and corrected.
- B) LSFs provide a higher signal-to-noise ratio than direct quantization of LPC coefficients.
- C) LSFs require fewer bits to represent because they are always positive integers.
- D) LSFs directly represent the fundamental pitch frequency, simplifying the excitation estimation.

Comment: LSFs are used because they offer a mathematically robust way to ensure the synthesis filter remains stable; if quantization causes roots to shift or lose their interlacing property, it can be easily corrected at the decoder.

2. Regarding the Opus audio codec, what is the primary technical advantage of its hybrid design?

- A) It is based solely on a fixed-rate pulse code modulation that simplifies synchronization in packet-switched networks.
- B) It dynamically switches or combines the Linear Prediction (LPC) engine and the Modified Discrete Cosine Transform (MDCT) engine to handle both speech and music efficiently across a wide range of bitrates.
- C) It relies exclusively on a psychoacoustic model identical to MP3, but with higher sampling frequencies.
- D) It avoids all forms of entropy coding to minimize decoding latency in real-time streaming applications.

3. Which of the following best describes current research trends in the future of multimedia audio coding?

- A) Abandoning lossy compression in favor of universal lossless compression for all real-time mobile communications.
 - B) Focusing exclusively on increasing the bit-depth of PCM signals to 64-bit to eliminate all quantization noise.
 - C) Moving from signal-processing-based models toward deep-learning-based end-to-end neural codecs that learn optimal representations directly from massive datasets.
 - D) Completely replacing packet-based transmission protocols with dedicated circuit-switched channels to ensure zero jitter.
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Transform Coding and JPEG

Open-Ended Questions

1. Explain the role of the Geometric Mean of variances in transform coding and how it relates to the coding gain.

Answer: In block coding with optimal resource allocation (e.g., using the Huang-Schulteiss formula), the resulting quantization distortion is proportional to the geometric mean of the variances of the transformed coefficients. The coding gain G_T is defined as the ratio between the distortion of direct sample-by-sample quantization and the distortion achieved by transform coding. Since orthogonal transforms preserve the arithmetic mean of the variances (energy conservation) but alter their distribution, a transform that achieves high energy compaction, thereby minimizing the geometric mean of the variances, maximizes the coding gain.

2. Describe the problem of “frequency leakage” in the Discrete Fourier Transform (DFT) when applied to signal compression and how the Discrete Cosine Transform (DCT) addresses it.

3. Explain the entropy coding process for AC coefficients in the JPEG standard and the significance of the “End of Block” (EOB) symbol.

4. Draw the scheme and describe the functional blocks of a JPEG encoder.

5. Compare the block-based DCT approach used in JPEG with the wavelet-based decomposition used in JPEG2000, specifically in terms of how they handle the image signal and the resulting artifacts.

Answer: JPEG uses an 8x8 block-based DCT, which processes the image in independent segments. While computationally efficient, this approach often leads to “blocking artifacts” at low bitrates because discontinuities at the block boundaries become visible. In contrast, JPEG2000 utilizes the Discrete Wavelet Transform (DWT), which performs a multiresolution analysis of the entire image (or large tiles). By applying the transform across the image, JPEG2000 avoids the rigid boundaries of JPEG. As a result, instead of blocking, JPEG2000 tends to exhibit “ringing” artifacts (blurring near sharp edges) at very low bitrates, providing a more visually pleasing subjective quality compared to traditional block-based methods.

6. Explain the fundamental shift in Rate-Distortion (R-D) optimization from classical codecs (e.g., JPEG) to neural compression methods.

Answer: Classical codecs rely on combinatorial selection, where the optimization process searches through a pre-defined, discrete set of tools (like fixed transform basis functions

and quantization step sizes) to minimize distortion at a given bit-rate. In contrast, neural compression uses continuous learning. It treats optimization as a design process in a continuous space, utilizing gradient descent to physically shape the network's filter weights by minimizing a differentiable loss function $\mathcal{L} = D + \lambda R$. Consequently, the transform in neural compression is not fixed but is “born” from the data and specifically tailored to the targeted bit-rate.

7. Describe the JPEG lossless coding process applied to the following table of quantized DCT coefficients:

10	3	0	0	0	0	0	0
-2	1	0	1	0	0	0	0
1	0	0	0	0	0	0	0
0	4	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Multiple-Choice Questions

1. What is the primary purpose of an orthogonal transform in the context of the transform coding paradigm?

- A) To sparsify the signal by concentrating energy into a small number of large coefficients.
- B) To increase the variance of the signal to improve quantization precision.
- C) To make the signal components statistically dependent.
- D) To reduce the total energy of the signal before quantization.

2. In the context of JPEG compression, what is the role of the quantization table?

- A) It is used to calculate the inverse DCT at the decoder.
- B) It is a fixed, mandatory table specified by the JPEG standard to ensure quality.
- C) It performs the actual frequency analysis of the image blocks.
- D) It defines the rate-distortion trade-off by determining the resolution for each DCT coefficient.

3. Which statement regarding the relationship between the Arithmetic Mean (AM) and Geometric Mean (GM) of variances in transform coding is correct?

- A) Any orthogonal transform preserves the AM of variances but can alter the GM of variances.
- B) Any orthogonal transform alters both the AM and GM of variances.
- C) Any orthogonal transform preserves the GM of variances but alters the AM of variances.
- D) The AM of variances is always minimized by the Karhunen-Loève Transform.

Correct answer: A. Comment: Because orthogonal transforms are isometries, they preserve the L^2 norm, which implies that the sum of the variances (related to the AM) remains constant, while the transform aims to minimize the GM to reduce distortion.

4. What is a primary advantage of the hierarchical (multiresolution) decomposition offered by the Wavelet Transform in JPEG2000?

- A) It eliminates the need for any form of entropy coding.
- B) It allows for progressive transmission and scalable reconstruction of the image at different resolutions and quality levels.
- C) It restricts the image analysis to only high-frequency components to save space.
- D) It forces the use of fixed 8x8 blocks, similar to the legacy JPEG standard.

6. Which of the following is a key reason why JPEG2000 is generally more efficient than JPEG?

- A) It reduces the image size by simply deleting high-frequency coefficients without quantization.
- B) It uses the same 8x8 DCT but applies it multiple times to the same data.
- C) It relies exclusively on loss-less compression, making it impossible to lose image quality.
- D) It avoids blocking artifacts by using a global or tile-based wavelet transform instead of small independent blocks, and uses very efficient bit-plane lossless coding and RD optimized truncation

7. What is the primary purpose of adding Additive Uniform Noise during the training phase of a neural codec?

- A) To make the non-differentiable quantization operation differentiable, enabling backpropagation through the quantizer.
- B) To increase the spatial resolution of the reconstructed image.
- C) To force the neural network to learn to filter out high-frequency noise from the original image.
- D) To simulate the effect of transmission channel errors on the latent variables.

Comment: Quantization is a “staircase” function with zero derivative almost everywhere, which blocks backpropagation. Adding uniform noise ($\mathcal{U}(-0.5,0.5)$) acts as a continuous relaxation, allowing gradients to flow during training.

8. Why do Convolutional Neural Networks (CNNs) outperform Multi-Layer Perceptrons (MLPs) when applied to image compression?

- A) CNNs utilize local connectivity and weight sharing, which drastically reduces the parameter count and enforces translation invariance.
- B) MLPs are too computationally simple and cannot perform non-linear activation.
- C) CNNs are inherently linear, making them more compatible with traditional transform coding standards like JPEG.
- D) MLPs require significantly more data to train than CNNs for the same image size.

Comment: MLPs treat images as flattened vectors, ignoring spatial topology and causing a “parameter explosion.” CNNs exploit local spatial correlations using sliding filters, which is much more efficient for image data.

Lossless coding principles

Open-Ended Questions

- 1. Explain the difference between Fixed-Length Coding (FLC) and Variable-Length Coding (VLC), and describe why VLC is theoretically superior for non-equiprobable sources.**
- 2. Discuss the importance of the prefix condition in Variable-Length Coding and how it relates to the concept of instantaneous decodability.**
- 3. Explain the mechanism behind Arithmetic Coding and why it is often preferred over Huffman coding in practical, high-performance applications.**

Answer: Arithmetic coding maps an entire input message sequence to a single fractional number within the range $[0,1]$. The range is recursively subdivided based on the probability of each incoming symbol. This approach allows to implement block coding of the input message, without having to generate the full, exponentially-complex dictionary as Huffman would do. Its overhead is larger than Huffman's (2 bits per block instead of one), but, as the block size grows large, it becomes negligible.

- 4. What are the two distinct mechanisms by which “block coding” improves the efficiency of lossless compression?**

Answer: Block coding improves performance through two primary contributions. First, for sources with dependencies, the joint entropy of a block $H(X^K)$ is strictly less than the sum of individual marginal entropies $\sum H(X_i)$, meaning that encoding blocks allows the system to exploit inter-symbol correlations. Second, even for independent variables, block coding addresses the “1-bit penalty” inherent in non-dyadic probability distributions; by grouping K symbols, the overhead of rounding codeword lengths to integer bits is distributed across the entire block, effectively becoming $1/K$ per symbol and vanishing as K approaches infinity.

5. Provide a synthetic comparison between the main lossless coding techniques (Exp-Golomb, Huffman, Arithmetic, Dictionary, Neural) in terms of complexity and Latency, and provide a typical use case for each of them

6. Describe the principle of the Huffman coding. For the following probability distribution, compute the optimal lossless code, and compare its average length to the source's entropy.

Symbol	A	B	C	D	E	F
Probability	35%	10%	7%	8%	12%	28%

Multiple-Choice Questions

1. According to Kraft's inequality, which condition must be satisfied for an instantaneous (prefix) code to exist with lengths $\{l_1, l_2, \dots, l_M\}$?

- A) $\sum_{i=1}^M 2^{-l_i} \leq 1$
- B) $\sum_{i=1}^M 2^{-l_i} > 1$
- C) $\sum_{i=1}^M l_i \leq 1$
- D) $\prod_{i=1}^M 2^{-l_i} \leq 1$

Comment: Kraft's inequality provides the necessary and sufficient condition for the existence of a prefix code. If the equality holds, the code is complete (or maximal), meaning no more codewords can be added without violating the prefix property.

2. Which of the following statements best describes the behavior of the Shannon entropy $H(X)$ for a binary random variable with probability p ?

- A) The entropy is independent of the probability p .
- B) The entropy is maximized when $p = 0$ or $p = 1$.
- C) The entropy is minimized when $p = 0.5$.
- D) The entropy is maximized when $p = 0.5$ and vanishes as $p \rightarrow 0$ or $p \rightarrow 1$

3. Why is Lempel-Ziv (e.g., LZW) considered a "universal" coding algorithm?

- A) It does not require prior knowledge of the source statistics or symbol probabilities.
 - B) It produces fixed-length codewords for all input sequences.
 - C) It is guaranteed to be optimal for all finite-length sequences.
 - D) It uses a static dictionary that is hardcoded for all possible inputs.
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Questions on Multimedia Quality Evaluation

Open-Ended Questions

1. Explain the difference between subjective and objective quality evaluation in multimedia systems and why both are necessary.

Answer: Subjective evaluation involves human participants who rate content based on their personal experience (Quality of Experience - QoE), which is considered the “gold standard” because it directly reflects human perception. However, it is time-consuming, expensive, and not suitable for real-time applications. Objective evaluation uses mathematical models and algorithms to predict quality scores (Quality of Service - QoS). Objective metrics are fast and repeatable, making them ideal for monitoring and optimization, but they may not perfectly correlate with human visual or auditory perception in all scenarios. Therefore, objective metrics are often validated against subjective test data to ensure reliability.

2. What are the key stages involved in designing a subjective quality test according to standardized guidelines?

3. Describe the main categories of objective quality metrics based on the availability of the original “reference” signal.

Multiple-Choice Questions

1. In the context of subjective testing, what is the main purpose of a “screening” phase for participants?

- A) To ensure that participants have basic visual or auditory acuity necessary to judge the content accurately.
- B) To teach participants how to use specific software tools for editing video.
- C) To determine which participants are the most expert in video compression standards.
- D) To select participants based solely on their age and educational background.

Correct answer: A. Comment: To guarantee the validity of subjective results, it is essential that subjects can perceive the stimuli correctly; screening ensures they meet the minimum sensory requirements defined by standards.

2. Which of the following best describes the “Full-Reference” (FR) objective quality assessment approach?

- A) It requires the original, undistorted signal to calculate the quality score by comparing it with the processed signal.
- B) It evaluates quality by looking only at the final compressed bitstream without any reference.
- C) It uses artificial intelligence to hallucinate what the original signal might have looked like.
- D) It relies entirely on the subjective opinion of the person watching the video.

3. Why is statistical analysis a critical component of subjective quality evaluation?

- A) To automatically convert subjective ratings into bit-rate requirements for the codec.
 - B) To allow the experimenters to ignore any data that does not support their initial hypothesis.
 - C) To remove outliers, account for inter-subject variability, and ensure that the mean opinion scores (MOS) are statistically significant.
 - D) To reduce the number of test participants required to zero.
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Scalar and Predictive Quantization

Open-Ended Questions

1. Explain the difference between a “mid-tread” and a “mid-rise” quantizer in the context of uniform quantization for signed data.

Answer: A mid-tread quantizer is characterized by having zero as one of its quantization levels. This is particularly advantageous for signals with small fluctuations around zero, as these fluctuations are mapped exactly to zero, effectively suppressing noise. Conversely, a mid-rise quantizer has zero as a threshold between two quantization levels. As a result, even very small noise around zero is mapped to either a positive or negative quantization level, which can lead to the amplification of low-level background noise in the quantized signal.

2. Define the concept of a “deadzone” in a quantizer and explain why it is frequently employed in lossy compression systems.

3. Why is scalar quantization alone often considered insufficient for effective compression of non-sparse data?

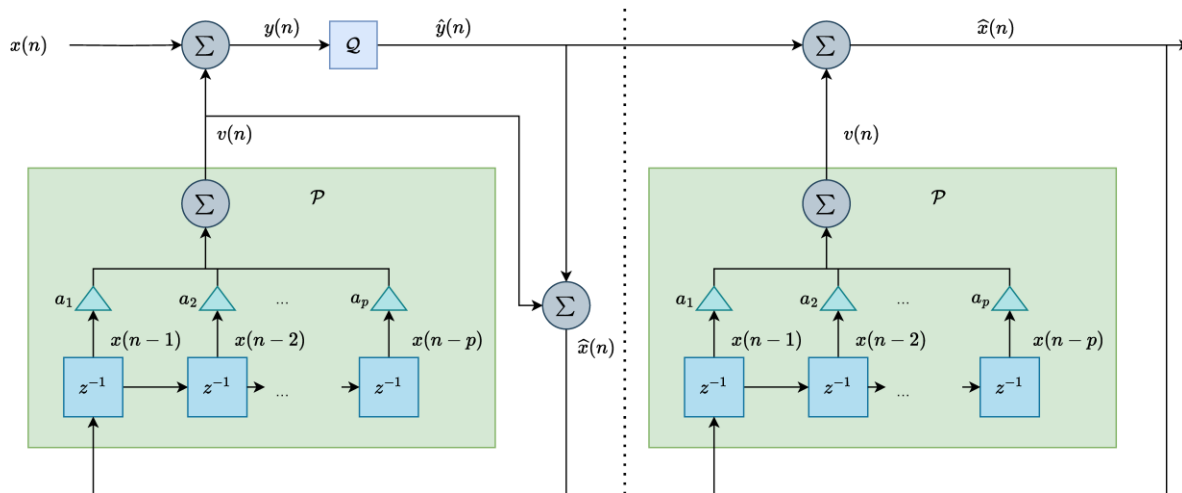
4. What is the condition for a predictive quantization system to be effective, and how is the “coding gain” defined?

Answer: Predictive quantization is effective if and only if the resulting prediction error has a smaller variance than the original signal. The coding gain (or prediction gain, G_p) is defined as the ratio between the variance of the original signal σ_X^2 and the variance of the prediction error σ_Y^2 , expressed in decibels: $G_p = 10\log_{10} \left(\frac{\sigma_X^2}{\sigma_Y^2} \right)$.

A positive coding gain indicates that the prediction successfully reduced the signal's energy, thereby improving the overall SNR for a given bit rate.

5. Draw the scheme of a linear predictive quantization scheme, and motivate the structure, with particular attention to the use of a decoding loop at the encoder side

The predictor $\mathcal{P}$ must be fed with the same data at the encoder and the decoder, otherwise there will be a drift. Thus, prediction is used to compute the quantized value $\hat{x}(n)$, which in turn is fed to the linear predictor.



Multiple Choice Questions

1. What is the primary purpose of the predictor in a predictive quantization system?

- A) To reduce the signal's variance and increase its sparsity by exploiting sample correlations.
- B) To increase the signal's dynamic range to match the quantizer levels.
- C) To perform transform coding on the entire block of data.
- D) To eliminate quantization noise completely from the system.

Comment: The core objective of predictive quantization is to leverage the correlation between neighboring samples to create a residual signal that is easier to compress than the raw input.

2. In a predictive quantization system, if the prediction $v(n)$ is nearly equal to $x(n)$, what happens to the variance of the signal $y(n)$ being sent to the quantizer?

- A) The variance of $y(n)$ becomes very small compared to the variance of $x(n)$.
- B) The variance of $y(n)$ remains identical to the variance of $x(n)$.
- C) The variance of $y(n)$ increases significantly.
- D) The variance of $y(n)$ becomes undefined.

3. When selecting a linear predictor of order P for a random process, how does the prediction error variance typically behave as the order increases?

- A) The benefit of including distant pixels diminishes rapidly due to the "screening effect," where close neighbors already capture most of the relevant information.
- B) The prediction error variance increases linearly with the order of the predictor.

- C) Higher-order predictors are always necessary to achieve any measurable coding gain.
- D) The order of the predictor has no impact on the variance of the prediction error.

Comment: While adding more past samples can improve prediction, there are diminishing returns because the immediate neighbors are the most highly correlated, making distant samples redundant in typical image or signal statistics.

4. In high-resolution uniform quantization, what is the approximate relationship between the SNR and the bit rate (R)?

- A) The SNR increases by approximately 6 dB for every additional bit of rate.
- B) The SNR decreases linearly as the rate increases.
- C) The SNR remains constant regardless of the number of quantization levels.
- D) The SNR increases exponentially with the rate (R^2).

Adaptive Streaming Systems

Open-Ended Questions

1. **Describe the fundamental architectural differences between a “Push-based” streaming system (e.g., RTP/UDP) and a “Pull-based” system (e.g., DASH).**

Answer: In push-based architectures, the server dictates the data flow, sending packets to the client according to a server-controlled schedule. This typically requires specialized transport mechanisms and can be problematic for network traversal. Conversely, pull-based systems shift control to the client, which autonomously requests media segments via standard HTTP. This allows the system to utilize standard web infrastructure, such as Content Delivery Networks (CDNs), and enables the client to adapt its requests based on local monitoring of network throughput and buffer state.

2. **Analyze the role of the client-side buffer in the context of stability and Quality of Experience (QoE).**
3. **Define “Switching Penalty” and discuss its impact on perceived video quality.**
4. **Explain the evolution of the playout buffer level $B(t)$ using a mathematical model. In your explanation, describe the dynamics of the “playback” (draining) phase and the “rebufferization” (stalling) phase.**

Answer: The dynamics of the buffer $B(t)$ are modeled as a continuous-time differential equation: $\frac{dB(t)}{dt} = C(t) - R(t)$, where $C(t)$ is the download rate (throughput) and $R(t)$ is the consumption rate (playback bit-rate).

Playback Phase ($B(t) > 0$): During normal operation, the system consumes data at rate $R(t)$. If the download rate $C(t)$ exceeds $R(t)$, the buffer fills. If $C(t) < R(t)$, the buffer drains. The video playback proceeds continuously as long as the buffer level remains above zero.

Rebufferization Phase ($B(t) = 0$): When the buffer level reaches zero, playback must stop to avoid an underflow error. The system enters a stall state where $R(t) = 0$ and the buffer level is forced to remain at zero until the accumulated downloaded data exceeds the playback requirement again. The time spent in this state is known as the “rebuffering time,” which is a primary negative factor for the user’s Quality of Experience (QoE).

Multiple Choice Questions

1. **What is the primary motivation for using HTTP-based protocols for video streaming?**

- A. The ability to bypass firewalls and NAT configurations using standard web ports (80/443).
- B. The theoretical capacity to provide sub-millisecond latency for real-time interaction.
- C. The requirement that the transmission bandwidth must be exclusively reserved.
- D. The elimination of the need for any client-side buffering mechanism.

Comment: Standard HTTP compatibility is the defining feature of DASH, ensuring that streaming traffic can traverse virtually all enterprise and residential network barriers without specialized hardware.

2. **What is the consequence of an ABR algorithm that systematically overestimates the available bandwidth?**

- A) The client requests segments that the network cannot deliver in time, resulting in buffer depletion and playback stalls.
- B) The server automatically increases the network capacity of the link.
- C) The video quality improves without any impact on network stability.
- D) The MPD file is dynamically rewritten to reduce the segment size.

Comment: Overestimating throughput causes the client to request data chunks that exceed the current network capability, directly leading to buffer starvation.

3. **Which metric is a direct indicator of streaming QoE from the end-user’s perspective?**

- A) The frequency and total duration of playback interruptions (rebufferization events).
- B) The total number of IP packets lost during transmission.
- C) The maximum bit-rate supported by the server infrastructure.
- D) The specific browser or application software version used for playback.

Comment: Users evaluate streaming quality primarily through the lens of continuous, uninterrupted viewing and perceptual stability, rather than low-level networking metrics like packet loss or hardware versions.

4. **At what stage in the session lifecycle does a DASH client process the Media Presentation Description (MPD)?**

- A) At the very beginning of the session, to acquire the map of available representations.
- B) After every single segment is successfully rendered to the screen.
- C) Only when the network connection is completely lost.
- D) Upon request by the authentication server.

Comment: The MPD functions as the foundational manifest of the content; the client must ingest this metadata initially to identify the available variants and determine how to construct the requests for specific media segments.

Motion Estimation e Video Coding

Part 1: Motion Estimation

Open-Ended Questions

1. **Describe the difference between “motion field” and “optical flow”.**
2. **Explain the Horn and Schunck algorithm’s core principle for dense optical flow estimation.**

Answer: The algorithm treats optical flow estimation as a global optimization problem. It combines two terms: a data attachment term, which minimizes the residual of the optical flow equation ($uf_x + vf_y + f_t = 0$), and a regularization term that penalizes spatial variations in the velocity field ($\nabla u^2 + \nabla v^2$). By using a Lagrange multiplier, it forces the flow field to be smooth, effectively resolving the flow estimation problem. In practical applications, the images are spatially sampled and the HS algorithm iteratively converges toward the solution

3. **Discuss the Rate-Distortion trade-off when selecting block sizes in motion estimation.**

Multiple Choice Questions

1. **Which of the following is a disadvantage of using the Sum of Squared Differences (SSD) as a matching criterion?**
 - A) It is sensitive to outliers and involves multiplications, increasing complexity.
 - B) It fails to calculate the energy of the prediction error correctly.
 - C) It produces a more regular motion vector field than SAD.
 - D) It is impossible to use in iterative search algorithms.

Comment: SSD squares the prediction errors, which heavily weights large outliers (e.g., pixels violating the constant illumination hypothesis), often leading to irregular motion vectors compared to SAD.

2. **What is the main benefit of the Hexagon Search strategy compared to Full Search?**
 - A) It guarantees finding the global minimum of the cost function in all cases.
 - B) It is designed specifically for affine motion models, not translational ones.
 - C) It eliminates the need for sub-pixel interpolation.
 - D) It significantly reduces the number of tested vectors while maintaining close-to-optimal performance.

3. **What does an affine motion model allow that a pure translational model does not?**
 - A) It uses fewer parameters, making it more robust to noise.
 - B) It can represent rotation, zoom, and shear using six parameters
 - C) It forces the motion vector to be the same for all pixels in an image.
 - D) It calculates motion without requiring any reference image.

Part 2: Video Coding Principles

Open-Ended Questions

1. **Why is the temporal prediction error usually more efficient to encode than the original video signal?**
2. **Describe the function of the “Mode Selection” step in a hybrid video encoder.**
3. **How does the “Channel Buffer” controller manage the trade-off between target rate and video quality?**

Answer: The controller monitors the buffer occupancy. If the buffer level exceeds a high threshold (indicating bits are being produced faster than they can be sent), it increases the quantization step size to lower the bit-rate. Conversely, if the buffer falls below a low threshold, it decreases the quantization step size to improve quality by using more bits.

Multiple Choice Questions

1. **What is the primary role of an “I-frame” in a GOP structure?**
 - A) **To provide a random access point and terminate temporal error propagation.**
 - B) To provide the highest compression ratio by using spatial prediction.
 - C) To store only the motion vectors, omitting the pixel data.

- D) To serve as a temporary buffer for B-frames.

Comment: I-frames are coded independently (Intra-coded), which makes them the only frames that can be decoded without reference to others, acting as anchors.

2. In the context of motion vector coding, why is a Median Predictor (MVP) used?

A) To exploit spatial correlation between adjacent motion vectors and reduce MVD bit-rate.

- B) To calculate the motion of the background more accurately than the foreground.
- C) To force all motion vectors in the image to be identical.
- D) To increase the complexity of the encoder to prevent overfitting.

Comment: Since motion in a scene is usually continuous, adjacent blocks have similar vectors. By predicting the current vector from neighbors (A, B, C), we only need to encode the difference (MVD), which is typically small and sparse.

3. What happens in the decoder when it receives an “Inter-coded” block?

A) It decodes the motion vector and prediction error, then adds them to the block from the reference frame.

- B) It immediately discards the frame and requests a re-transmission.
- C) It performs a new motion estimation search using the current block.
- D) It uses only spatial filtering to reconstruct the image.

Comment: In Inter-mode, the decoder relies on the motion information provided in the bitstream to find the predictor in the frame buffer and adds the decoded residual to it.

Part 3: Modern Video Compression Standards

Open-Ended Questions

- 1. What is the specific scope of video compression standards like H.266/VVC?**
- 2. Explain the advantage of the “Coding Tree Unit” (CTU) structure introduced in HEVC/VVC.**

Answer: Unlike fixed-size macroblocks, CTUs allow for flexible, recursive partitioning (Quad-tree and Multi-type trees). This enables the encoder to use very large blocks (up to 128x128) for homogeneous areas like sky or walls, and fine-grained, small blocks for detailed textures or object boundaries, significantly improving R-D efficiency in high-resolution video.

- 3. What are the roles of VCL and NAL in modern video standards?**

Multiple Choice Questions

- 1. What is the main purpose of the CABAC entropy coder in modern standards?**

- A) To adaptively assign shorter bit sequences to more frequent symbols based on context.
- B) To convert the image into the frequency domain.
- C) To replace motion compensation with a simpler technique.
- D) To increase the bit-rate by adding redundancy for error correction.

2. Why are “Tiles” considered “hardware-friendly” in VVC and HEVC?

- A) They allow independent processing of rectangular regions, enabling parallel encoding and decoding.
- B) They provide better visual quality for moving objects.
- C) They remove the need for entropy coding.
- D) They force the encoder to use only one core.

Comment: Tiles restrict motion vectors and prediction dependencies to specific boundaries, allowing multiple CPU or GPU cores to process different parts of the same frame simultaneously.

3. What is the function of an “In-Loop Filter” like the Adaptive Loop Filter (ALF)?

- A) To reduce the quantization artifacts and restore details at the reconstruction stage.
- B) To speed up the motion estimation process by simplifying the image.
- C) To generate random grain noise to hide poor lighting.
- D) To replace the transform block in the encoder.

Comment: Since quantization is lossy and block-based, it creates artifacts like blocking or ringing. Filters inside the reconstruction loop ensure these artifacts are smoothed before the frame is stored in the buffer as a reference for future frames.
